Dirac Notes for Integration over Energy Surfaces

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1 General Concept

The Dirac delta distribution $\delta(x)$ is defined by its action on a sufficiently smooth test function $f(x)$ under an integral:

$$\int_{-\infty}^{\infty} f(x)\delta(x)\,dx = f(0).$$

2 Shifting Properties

The Dirac delta can be seen as an operator that "picks out" the value of a function at a specific point. For a function $f(x)$ and a point a, we consider:

$$\int_{-\infty}^{\infty} f(x)\delta(x-a)\,dx.$$

By substituting $u = x - a$, we have $dx = du$. As $x \to \pm\infty$, u also goes to $\pm\infty$. The integral becomes:

$$\int_{-\infty}^{\infty} f(u+a)\delta(u)\,du = f(0+a) = f(a).$$

This result is fundamental and is often referred to as the **sifting** or **sampling property**.

3 Composition with Functions

3.1 Linear Scaling

Consider a linear scaling of the argument ax within the delta function:

$$\int_{-\infty}^{\infty} f(x)\delta(ax)\,dx.$$

Substituting $u = ax$, which implies $dx = \frac{du}{a}$, we have:

$$\int_{-\infty}^{\infty} f(x)\delta(ax)\,dx = \int_{-\infty}^{\infty} f\left(\frac{u}{a}\right)\delta(u)\frac{du}{a}.$$

Since the Dirac delta only evaluates the function at $u = 0$, it fulfills the symmetry property $\delta(u) = \delta(-u)$. This ensures that the result depends only on the magnitude of the scaling factor, yielding:

$$\int_{-\infty}^{\infty} f(x)\delta(ax)\,dx = \frac{1}{|a|}f(0).$$

3.2 Composition with a General Function

If the argument is a non-linear function $g(x)$, the delta function triggers only at the roots x_i where $g(x_i) = 0$. Assuming first a single root x_0, we linearize $g(x)$ around this point using a Taylor expansion: $g(x) \approx g'(x_0)(x - x_0)$. Using the scaling rule from above, we find:

$$\int_{-\infty}^{\infty} f(x)\delta(g(x))\,dx \approx \int_{-\infty}^{\infty} f(x)\delta(g'(x_0)(x - x_0))\,dx = \frac{1}{|g'(x_0)|} f(x_0).$$

If $g(x)$ has multiple isolated roots x_i, the integral generalizes to a sum over all these points:

$$\int_{-\infty}^{\infty} f(x)\delta(g(x))\,dx = \sum_i \frac{f(x_i)}{|g'(x_i)|}.$$

3.3 Integration over Energy Surfaces

The dispersion relation $E(\mathbf{k})$ defines energy surfaces in momentum space. To integrate a function $f(\mathbf{k})$ over a constant energy surface E_0, we use the delta function:

$$\int f(\mathbf{k})\delta(E(\mathbf{k}) - E_0)\,d\mathbf{k}.$$

The delta function restricts the integration to points where $E(\mathbf{k}) = E_0$. The increase of the energy is given locally by its gradient $\nabla_{\mathbf{k}} E(\mathbf{k})$. A Taylor expansion around a point $\mathbf{k}_0$ on the energy surface yields:

$$E(\mathbf{k}) \approx E(\mathbf{k}_0) + \nabla_{\mathbf{k}} E(\mathbf{k}_0) \cdot (\mathbf{k} - \mathbf{k}_0).$$

Only the component of $\mathbf{k}$ parallel to the gradient (normal to the surface) contributes to the shift in energy. Applying the scaling property from the 1D case to this normal direction, we find:

$$\int f(\mathbf{k})\delta(E(\mathbf{k}) - E_0)\,d\mathbf{k} = \int_{E(\mathbf{k})=E_0} \frac{f(\mathbf{k})}{|\nabla_{\mathbf{k}} E(\mathbf{k})|}\,dS,$$

where dS is the surface element on the energy surface defined by $E(\mathbf{k}) = E_0$.